

What the Model Says About the Twelve Steps and the Twelve Traditions

A reference primer for *Nobody in Charge*

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How to read this

This is a reference document, not a chapter. It gives one short finding for each Step and each Tradition, and it exists so that a reader who wants to know what the model says about a particular item does not have to reconstruct it from six chapters. Everything here is drawn from the manuscript and is reproducible from `model/book-calculations.ipynb`. Nothing here is new.

Every entry comes in two parts. **Technical** states the result and its provenance. **In plain terms** says what it would mean to somebody sitting in a room, and says so without arithmetic. The second part is an interpretation of the first and is softer than it. Where the plain reading would carry further than the technical one supports, the entry says where it stops.

Three different instruments produce the findings below, and they are not interchangeable. Every entry says which one it used.

Algebra. Exact calculation on two fixed matrices. The consumption matrix S is twelve Steps by eight group resources. The governance matrix G is twelve Traditions by the same eight resources. Their product

$$B = S G'$$

gives the demand each Step places on resources each Tradition governs. These figures carry no sampling error. They are also no better than the two matrices, both of which one person built, and the largest outstanding item in the whole project is a second reader marking which of the ninety-six governance cells are non-zero.

Deliberation. Closed-form results from Golub and Jackson's 2010 theorem on naive learning. These concern how a room's consensus tracks the truth as the room grows. They are exact and they involve no simulation.

Simulation. Monte Carlo runs of the group model. Every figure quoted here comes from at least 400 seeds and carries an interval.

Four standing cautions

One. The coupling is not used by the simulation.

Technical. $B = S G'$ is a derived object. No entry of it was chosen and no entry of it feeds a run. It is a description of the two matrices, and it inherits every judgement in them.

In plain terms. The table matching Steps to Traditions was not built by watching groups. It falls out of two lists somebody wrote down: what each Step needs from other people, and what each Tradition looks after. If those two lists are wrong, everything in Part Four is wrong with them, and no amount of computing will reveal it.

Two. At full adherence the governance matrix cancels exactly.

Technical. It is column-normalised, so when every Tradition is at 1.0 the governance quality of every resource is identically 1. Thirty-five of the model's 118 hand-chosen numbers cannot affect a fully adherent group at all.

In plain terms. In a group doing everything right, the Traditions do not show up in the numbers, because there is nothing left for them to fix. Everything the model says about individual Traditions is a statement about groups falling short somewhere, which is every real group. The Traditions are visible only in the breach.

Three. Only two simulation claims survive full sensitivity testing.

Technical. Across all 118 parameters, varied globally and one at a time over 236 targeted perturbations, exactly two results hold: that a fully adherent group persists, and that losing referrals is worse than losing attraction in the sense of being more likely to kill the group. Read as “ends up smaller” that second ordering reverses under three of four changes to the model's architecture.

In plain terms. Two conclusions here are solid enough to argue with somebody about. A group that keeps all twelve Traditions keeps going, and a group cut off from outside referrals is in more danger than a group nobody has heard of. Everything else in the simulation is a shape rather than a price. If a figure below says a group loses sixty-six per cent of something, read the direction and the fact that it is large. Do not read the sixty-six.

Four. The degradation figures rank one outcome at one point.

Technical. Where a Tradition entry quotes a membership loss, the design is: all twelve Traditions at 0.85, then one Tradition alone lowered to 0.50, twenty-year horizon, 400 paired replications under common random numbers, reference membership 23.4 with a cross-seed standard deviation of 11.9.

In plain terms. That column answers one question: if a group let this Tradition slide while holding the others steady, how many fewer people would be in the room in twenty years. It is not a measure of how much a Tradition matters. It says nothing at all about whether the group's decisions are any good, which is where Traditions 2, 9 and 12 do most of their work. A Tradition can score near zero here and still be carrying one of the book's three headline results. One of them does.

Part One. The Twelve Steps

Each entry gives the derived group-dependence coefficient beta, which is the Step's row sum in S normalised by the largest row sum, and the Step's principal supplier, which is the Tradition

governing the largest share of what that Step consumes. The index-mate is the Tradition carrying the same number, and its rank among the twelve is the whole of Part Four's argument in one column.

Step One, admitting powerlessness

Technical. $\beta = 1.00$, the joint maximum. Principal supplier Tradition 3 at 1.22, with Tradition 1 second at 0.99. Its largest consumption is identification, others in the room who name themselves as alcoholic, at the matrix maximum of 1.0, followed by admission at 0.8. It carries the highest top speed of any Step, 0.30 per week, and the lowest exposure to the maintenance gate, 0.05. Its index-mate, Tradition 1, ranks second at 0.99 against the winner's 1.22, the closest any index-mate comes. Under the sixty-four resource-list variants Step One regains its index-mate in fifteen of them, more than any other Step. *Algebra, plus the simulation's parameter table.*

In plain terms. This is the Step you cannot take by yourself, and it is tied with the twelfth for depending most on other people. What it needs is not advice. It is other people in the room saying the same thing about themselves, and a door that opened without asking anything first. It is also the fastest Step to move and the one least affected by how much you already have to lose, which is the model's way of saying that an arrival has nothing yet to protect. It is the one place where the obvious pairing nearly works, and it is the Step most likely to pair with its own Tradition if somebody redrew the list of what groups supply.

Step Two, coming to believe

Technical. $\beta = 0.71$. Principal supplier Tradition 11 at 0.82, with Tradition 5 second at 0.72. Its dominant input is visible proof that recovery happens, at 1.0. The index-mate, Tradition 2, ranks seventh at 0.13. *Algebra.*

In plain terms. The model treats coming to believe as an evidential matter rather than a persuasive one. What the Step consumes is people visibly getting better in front of you. That is why the Tradition supplying it is attraction rather than promotion: nobody is talked into this, they watch it happen and draw the obvious conclusion. If the model is right about Step Two, then a group's most important argument for itself is the condition of the people in it.

Step Three, the decision

Technical. $\beta = 0.29$, near the bottom. Principal supplier Tradition 2 at 0.31, with Tradition 1 second at 0.27. Row sum 0.70, so it asks little of the group in absolute terms. Index-mate Tradition 3 ranks seventh at 0.01, the smallest non-zero index-mate entry in the matrix. *Algebra.*

In plain terms. The decision asks less of a group than almost any other Step. What it does draw on is counsel: somebody to talk it over with. The Tradition sharing its number contributes essentially nothing to it, and that is the starkest single number against the pairing idea anywhere in the matrix.

Step Four, the inventory

Technical. $\beta = 0.33$. Principal supplier Tradition 2 at 0.35, with Tradition 1 tied at 0.35 to the second decimal and behind by four parts in a thousand, the narrowest top-two margin in the matrix. Its largest consumption is gentle pressure at 0.4 and counsel at 0.3. The index-mate, Tradition 4,

has an entry of exactly zero, because Tradition 4 governs no resource any Step consumes. That zero is arithmetic and not evidence: no multiplicative perturbation can move it. The threshold test that can reach it finds that a uniform governance strength of 0.438 would be needed for the index-mate to win, against a mean live entry in the governance matrix of 0.374. *Algebra.*

In plain terms. The inventory needs two things from a group: mild expectation that you will actually do it, and somebody to ask about how. Tradition 4 supplies neither, and it supplies nothing to any other Step either, because it is one of the five Traditions in the model that guard rather than provide. So the pairing fails here for a boring reason rather than an interesting one, and the primer says so rather than counting it as a discovery.

Step Five, telling someone

Technical. $\beta = 0.71$. Principal supplier Tradition 12 at 1.08, with Tradition 1 second at 0.65, a margin of 0.43, the second widest in the matrix. Step Five is the only Step that consumes confidentiality at the matrix maximum of 1.0, and Tradition 12 is the Tradition that governs confidentiality at 1.0. Its robustness is asymmetric: it survives 99.5 per cent of draws at plus or minus thirty per cent jitter, with a 95 per cent Wilson interval of [99.1, 99.7] on 2,000 draws, and only 17.5 per cent [15.9, 19.2] when every non-zero magnitude is replaced at random. Its index-mate, Tradition 5, ranks fifth at 0.12. *Algebra.*

In plain terms. Telling someone requires that it stay told. Step Five is the only Step that needs confidentiality at full strength, and anonymity is the only Tradition that supplies it at full strength. That is the cleanest match in the book, and it is also the clearest illustration of what these matches rest on. If you accept my numbers it is near certain. If you accept only the pattern of which Tradition touches what, and let the strengths fall where they may, it mostly disappears. The pairing is a judgement argued for, not a result computed, and the difference between ninety-nine and seventeen is exactly the size of the judgement.

Step Six, becoming willing

Technical. $\beta = 0.25$. Principal supplier Tradition 1 at 0.25, with Tradition 2 second at 0.24, a margin of 0.01 and the narrowest of any row except Step Four's, which is a tie. Row sum 0.60. Index-mate Tradition 6 is a structural zero and ranks ninth. Its threshold is 0.417, the lowest of all twelve. *Algebra.*

In plain terms. Becoming willing barely needs a group at all: a little pressure, a little counsel, and that is most of it. The two Traditions at the top are so close that the model has no real opinion about which one carries it. Of all twelve Steps this is the one where the pairing comes nearest to being recoverable, and it still needs its own Tradition to look after its needs more attentively than a typical entry in the whole matrix.

Step Seven, asking

Technical. $\beta = 0.17$, the minimum. Row sum 0.40, the smallest in S. Principal supplier Tradition 1 at 0.17. The ratio of the most group-dependent Step to this one is 6.0. Index-mate Tradition 7 is a structural zero and ranks tenth. Setting a uniform governance strength at the mean live entry of 0.374 for all twelve Steps at once, Step Seven's index-mate loses by the narrowest margin of the twelve, 0.02. *Algebra.*

In plain terms. This is the most private Step in the programme. If you asked how much of each Step happens in a room rather than in a person, Step Seven gives the smallest answer, by a factor of six against the largest. That is not a claim that it is easy, or minor, or that it happens without the rest. It is a claim about how much of it other people can supply, and the answer the model gives is: almost none of it.

Step Eight, listing the harms

Technical. $\beta = 0.33$. Principal supplier Tradition 2 at 0.43, with Tradition 1 second at 0.29. Its largest consumption is counsel at 0.4. The index-mate, Tradition 8, ranks fourth at 0.07, the best rank achieved by any index-mate other than Step One's. *Algebra.*

In plain terms. Making the list is mostly a matter of having somebody to ask, which is why the group conscience supplies it. Its own numbered Tradition does better here than in most rows and still comes fourth, which is the pattern across the whole table: the pairing is not absurd anywhere, and it wins nowhere.

Step Nine, amends

Technical. $\beta = 0.62$. Principal supplier Tradition 2 at 0.90, with Tradition 1 at 0.48, a margin of 0.42 and the third widest in the matrix behind Step Ten's 0.51 and Step Five's 0.43. Step Nine consumes counsel at 0.9, the largest entry in S other than the four entries at 1.0. It carries the lowest top speed of any Step, 0.15 per week. Disabling the twelfth Step degrades Step Nine by 28 per cent at 400 seeds, even though Step Nine has no direct dependence on service; the degradation travels through maintenance capacity, of which Steps Ten to Twelve are the average. Index-mate Tradition 9 is a structural zero and ranks eleventh. *Algebra and simulation.*

In plain terms. Amends is the Step that most needs counsel, and it is the slowest Step to move, which between them describe something people already know: you do not do this one quickly and you do not do it without asking. The finding worth carrying is not about Step Nine's own needs. Switch off the twelfth Step, so nobody in the group is carrying the message to anyone, and Step Nine gets substantially worse, despite having nothing whatever to do with service. It gets worse because service is part of what keeps a person able to hold on to anything at all. That is what people mean when they say you keep it by giving it away, and here it is an output of the model rather than an assumption fed into it.

Step Ten, the daily inventory

Technical. $\beta = 0.62$. Principal supplier Tradition 1 at 0.94, with Tradition 2 second at 0.43, a margin of 0.51, the widest in the matrix. Its largest consumptions are gentle pressure at 0.7 and week-to-week continuity at 0.5, which between them are the two resources Tradition 1 supplies most heavily. Index-mate Tradition 10 is a structural zero and ranks twelfth, the worst rank in the table. *Algebra.*

In plain terms. Daily inventory is a habit, and habits need the two things a group supplies steadily rather than dramatically: the meeting happening again next week, and other people mildly expecting you at it. That is the whole of what the model means by unity here, and it is a duller thing than the word suggests. Step Ten also has the worst showing of any Step for the pairing idea. Its own numbered Tradition comes dead last of twelve.

Step Eleven, the conscious contact

Technical. $\beta = 0.33$. Principal supplier Tradition 1 at 0.47, with Tradition 2 second at 0.21. Index-mate Tradition 11 ranks fourth at 0.11. With Steps Ten and Twelve it forms the maintenance capacity term: the model computes a member's capacity to hold what they have as a Hill function of the mean of Steps Ten, Eleven and Twelve, and that capacity multiplies the growth of every Step, weighted by an exposure rising from 0.05 at Step One to 1.00 at Step Twelve. *Algebra and the simulation's structure.*

In plain terms. Its own row is unremarkable. Its importance is structural: along with Ten and Twelve it is what the model calls maintenance, the capacity to keep hold of what has already been gained. Those three set a multiplier on everything else, weighted so that it barely touches Step One and fully governs Step Twelve. In plainer language, the last three Steps are what stop the first nine leaking away, and the further along somebody is, the more the leak matters. A newcomer has nothing to maintain. A member of fifteen years has almost everything to maintain.

Step Twelve, carrying it

Technical. $\beta = 1.00$, the joint maximum, and the Step that consumes seven of the eight resources. Principal supplier Tradition 5 at 1.25, with Tradition 3 second at 1.10. It is the only consumer of the recipient resource, which is the only one of the eight not produced by the members themselves: supply is newcomers per available helper, so both a shortage of newcomers and a shortage of helpers reduce it. Three simulation results at 400 seeds, thirty years, full adherence. Setting the twelfth Step's growth rate to zero costs 66 per cent of membership, 41.7 plus or minus 1.5 falling to 14.2 plus or minus 0.4, and 16 per cent of average practice. It does not kill the group: survival is 400 of 400. Removing the twelfth Step's dependence on the recipient resource raises membership by 8.7 per cent. Index-mate Tradition 12 ranks sixth at 0.12, and this is the fragile row of Part Four: the top two are 1.25 and 1.10, a margin of 0.15, and the Step Twelve to Tradition 5 assignment survives only 67.3 per cent [65.3, 69.4] of draws at plus or minus thirty per cent jitter. *Algebra and simulation.*

In plain terms. Carrying the message is not a reward collected at the end. It holds the rest up. Turn it off and the group loses two thirds of its people and gets flatter across every Step, and yet it does not die. It settles into a smaller, quieter version of itself and stays there indefinitely, which is a shape a real group might not recognise as failure at all. The second finding is the uncomfortable one. What limits how much twelfth-step work an experienced member can do is the supply of newcomers, and if you remove that limit everything improves. The model is saying the group needs newcomers for the sake of the people who are already there, which is a less flattering account of welcome than the one usually given, and it follows from the arithmetic rather than from any view about motives.

What the Step rows say taken together

Technical. Group-dependence is highest at the two ends, 1.00 at Step One and Step Twelve, and lowest in the middle, 0.17 at Step Seven, with a mean of 0.53. That the entry Step and the service Step depend most on other people is a consequence of the resource assignment rather than an input to it. The transmission asymmetry follows: a member's contribution to the group carries weight 1 by construction, while the group's contribution back is multiplied by β , so the ratio runs from

1 to 6.0 and averages 1.9. The ordering claim, that a Step cannot be skipped, is a limiting case: writing the stage as a constant-elasticity-of-substitution aggregator, strict ordering holds for every substitution parameter ρ less than or equal to zero. ρ is estimable. With three proxies and 1,200 observations, 25 replications per cell, bias never exceeds 0.21 and the recoverable band around zero is roughly plus or minus 0.15.

In plain terms. The Steps form a U. The first and the last are things you cannot do alone. The middle ones are largely yours to do, with a group nearby rather than involved. Nobody designed that shape and it was not put in by hand; it comes out of asking, for each Step in turn, what it needs from other people.

The second point is about the rule that you cannot skip a Step. That has always been stated as a piece of folk wisdom, take it or leave it. The model shows it is not a separate belief at all. It is one setting of a dial that runs continuously from “each Step strictly requires the one before” to “the Steps substitute freely for each other”, and the folk rule is everything on one half of that dial rather than a single extreme point. Which means it is the kind of claim that could be measured. Somebody with the right data on twelve hundred people could put a number on how much the Steps really do depend on their order, and the model says roughly how precise that study would need to be.

Part Two. The Twelve Traditions

Each entry gives the Tradition’s load, which is its column sum in B and therefore the total demand all twelve Steps place on resources it governs, how many of the eight resources it governs, and its membership loss in the degradation comparison described in caution four. Losses whose 95 per cent interval includes zero are marked unresolved: eight of the twelve comparisons resolve and four do not.

Tradition 1, unity

Technical. Load 6.52, the largest, governing all eight resources and the only Tradition to do so. Principal supplier for four Steps and runner-up for six, so it is in the top two for ten of the twelve. Degrading it alone costs 4.48 members [3.18, 5.77], third largest. Its primacy survives 75.4 per cent [73.5, 77.2] of fully structural draws, against 40.6 per cent for the index-pairing claim. It is not the most diffuse Tradition by concentration: singleness of purpose scores marginally lower at 0.197 against unity’s 0.204. Its load is carried by two resources: continuity at 1.89 and gentle pressure at 1.86 are 57.5 per cent of the total, and transferring either one to Tradition 5 makes Tradition 5 the leader. Six of eight such transfers cannot flip it and two can. *Algebra, with simulation for the degradation figure.*

In plain terms. Unity is not one of the things a group supplies. It is the condition of everything a group supplies, which is why it is the only Tradition touching all eight and why it is in the top two for ten of the twelve Steps. But look at what that actually consists of, and most of it is two unglamorous things: the meeting keeps happening, and people notice whether you are at it. Take those two away and unity drops to third place. So the model’s “unity” is closer to reliability and mild social expectation than to fellow feeling, and a group worried about its unity would do better to check whether it has cancelled a meeting than to check whether everyone is getting on.

Tradition 2, the group conscience

Technical. Load 3.89, second largest, governing four resources, with counsel at 0.69 of its load and the highest concentration of any Tradition at 0.520. Principal supplier for four Steps and runner-up for four more. Degrading it alone costs 3.43 members [2.13, 4.72]. In the deliberation model it is one of three Traditions keeping maximum influence falling toward one over N , which is the Golub and Jackson condition for a consensus converging on the truth. A single member holding 0.35 of every row floors the group's error at 0.279 however large the group grows, against a flat error falling as one over the square root of N . Five per cent of every row leaves the error 1.84 times the flat benchmark at a thousand members, and the factor grows without bound. *Algebra, deliberation and simulation.*

In plain terms. In the simulation, the group conscience is mostly what supplies counsel: it is the Tradition behind there being somebody to ask. Its real work is somewhere the membership numbers cannot see it. It is what keeps a room deciding by adding up what everybody thinks rather than by deferring to one person, and the theorem behind that is unforgiving. A group that leans on one member is permanently worse at being right, and it does not matter how big the group gets, because the leaning does not dilute. Nor does the room have to be dominated for this to bite. A member who holds five per cent of everyone's attention, which is not much and would not look like a problem from inside, nearly doubles how wrong a large group ends up. The damage starts long before anybody would call it a problem.

Tradition 3, the open door

Technical. Load 2.69, governing four resources, with admission at 0.37 of its load. Degrading it alone costs 5.16 members [3.91, 6.40], second largest in the comparison. The figures that follow come from a different design and the two must not be added together: Tradition 3 varied alone across five levels with every other Tradition at 1.0, 400 seeds per level, thirty-year horizon. Because the Tradition removes the group's power to refuse admission, Tradition 3 does not appear in the arrival rate at all. It appears only in the dropout hazard, as a friction largest for members whose practice is near zero and negligible for veterans. An unwelcoming group's measured quality rises from 0.354 at full adherence to 0.400 at zero, and its newcomer share rises from 0.122 to 0.201. Both are conditional on survival and the quality advantage is smaller than it looks for that reason. Going to the unwelcoming extreme costs 14.2 members and 5.5 points of survival at thirty years. *Algebra and simulation.*

In plain terms. The open door is modelled as something a group cannot do rather than something it does. A group has no power to decide who turns up; it only decides how people are treated once they have. So the Tradition acts on who stays, not on who arrives. Two consequences fall out that nobody put in, and both are unsettling. An unwelcoming group looks *better* on paper, because it loses its shakiest members first and the average of who is left goes up. And it has more newcomers in the room at any given moment, because it has to keep replacing the people it drove off. So if you walk into a meeting with a lot of new faces and a lot of solid-looking long-timers, the model says that is either a healthy group or a cold one, and you cannot tell which from the room. That is the most testable claim in the book and the easiest to falsify: count the newcomers across meetings of differing warmth. If cold meetings have fewer newcomers, this mechanism is wrong.

Tradition 4, autonomy

Technical. Load 0.00. Its row in the governance matrix is identically zero: it governs no resource any Step consumes. It is one of the five protective Traditions and enters the simulation only as a multiplier, paired with Tradition 7, on the effective adherence of everything else. Degrading it alone costs 2.82 members [1.45, 4.19], identical to Tradition 7's to the last digit because the two enter the same term symmetrically. That identity is an artefact of the model's construction, not a finding about the Traditions. *Algebra and simulation.*

In plain terms. Autonomy hands the Steps nothing directly. Its whole job in the model is keeping the other Traditions from being overridden from outside, and it does that jointly with self-support. The two share a number because the model treats them as a pair, not because anybody discovered they were equally important. What *is* derived here is only that autonomy supplies nothing; how it should act instead was assumed, and the multiplier is the simplest assumption that gives it any role at all.

Tradition 5, singleness of purpose

Technical. Load 3.88, a hair behind the group conscience, governing six resources, the second broadest. Principal supplier for Step Twelve and runner-up for Step Two. Degrading it alone costs 2.27 members [0.94, 3.60]. Transferring unity's governance of continuity or of gentle pressure to singleness of purpose makes it the leader, which is the strongest form of the objection that later AA literature substituted unity for single-purposedness. Six of the eight possible transfers cannot flip it and two can. *Algebra and simulation.*

In plain terms. Singleness of purpose is unity's only real rival for the most load-bearing Tradition, and the whole contest comes down to who owns two things: the meeting continuing to happen, and the pull of other people expecting you there. If those belong to unity, unity leads. If they belong to the group having one job, single purpose leads. That is a real question about how AA works and the model cannot settle it. What the model can do is say that this, and nothing else in the matrix, is where the answer would change.

Tradition 6, no endorsement

Technical. Load 0.00, a protective Tradition with an empty governance row. Enters the simulation as a multiplier, paired with Tradition 10, on the effective adherence of singleness of purpose. Degrading it alone costs 0.71 members [-0.42, 1.83], **unresolved**: the comparison cannot distinguish its effect from zero at 400 paired replications. Identical to Tradition 10's figure for the same structural reason as the Tradition 4 and 7 pair. *Algebra and simulation.*

In plain terms. Refusing endorsements supplies nothing to any Step. Its job is protecting the group's single purpose from being diluted, alongside Tradition 10. The simulation cannot tell its effect apart from nothing at all, and the correct reading of that is that the simulation has nothing to say about it, not that it does nothing. An instrument that cannot resolve a thing is silent about it, not against it.

Tradition 7, self-support

Technical. Load 0.00, protective, empty row. Paired with Tradition 4 as a multiplier guarding against outside override. Degrading it alone costs 2.82 members [1.45, 4.19], identical to Tradition 4's by construction. Money enters the simulation elsewhere, through a solvency term scaling the

continuity resource: a group whose established members cannot cover the rent supplies continuity at 0.45 rather than 1.0. That mechanism is not attributed to Tradition 7 in the governance matrix. *Algebra and simulation.*

In plain terms. Passing the basket looks after nothing directly in the matrix, and yet money is unmistakably in the model somewhere else: a group that cannot pay its rent supplies week-to-week continuity at less than half strength, and continuity is one of the two things holding unity up. So the practical content of self-support is that the meeting keeps happening. The matrix does not give Tradition 7 credit for that, and a reader could reasonably say it should. This is one of the places a second person building the governance table would most likely disagree with the first.

Tradition 8, non-professional

Technical. Load 1.11, the smallest non-zero load, governing four resources with confidentiality the largest at 0.41 of the total. Principal supplier for no Step and runner-up for none. Degrading it alone costs 1.03 members [-0.18, 2.23], **unresolved**. *Algebra and simulation.*

In plain terms. This is the Tradition the model can say least about. It touches four things lightly, leads on none of them, and its cost in members cannot be told apart from zero. Stating that plainly is better than dressing it up. Keeping AA unpaid may matter a great deal for reasons this model was never built to see, and the honest report is that the instrument did not detect anything rather than that there is nothing there.

Tradition 9, no hierarchy

Technical. Load 0.00, protective, empty row. Degrading it alone costs 0.89 members [-0.26, 2.04], **unresolved**. In the deliberation model, rotation of service works only if the rotating pool scales with the group. A fixed pool floors maximum influence at roughly the officeholder's share divided by the pool size, while the flat benchmark keeps falling as one over N, so the gap grows without limit: a pool of twelve sits at 2.1 times the flat benchmark at fifty members and 23.9 times at eight hundred. The pool needed to stay within a factor of two of flat is 26 per cent of the group at every size tested from fifty to eight hundred. The 26 per cent depends on the parameter choices; the divergence does not. *Algebra, deliberation and simulation.*

In plain terms. This is the sharpest warning against reading the membership column as importance. On that column Tradition 9 does nothing measurable, and it carries one of the three headline results in the book.

Rotation is where the model knows something AA does not say. The Traditions say to rotate. They do not say how many people. The model says the answer is a fraction of the group and not a headcount, and that the difference is not a matter of degree. A group of fifty rotating twelve people through its service positions is fine. A group of eight hundred rotating twelve is a permanent oligarchy, whoever those twelve are and whatever anybody intends, and from inside it looks exactly like the healthy small group did. Roughly a quarter of the group needs to be in the pool. Treat the quarter as an order of magnitude rather than a target, and treat the underlying question as the durable one: if this group doubled, would the same people still be running it?

Tradition 10, no outside issues

Technical. Load 0.00, protective, empty row. Paired with Tradition 6 as a multiplier on singleness of purpose. Degrading it alone costs 0.71 members [-0.42, 1.83], **unresolved**, identical to Tradition

6's by construction. Part One's historical material bears on it more than the model does: the Washingtonians had a written analogue of this Tradition in print within two years of founding. *Algebra and simulation.*

In plain terms. Staying out of outside controversies supplies nothing directly and protects single purpose. The model cannot resolve its effect. The history is the more interesting evidence here, and it cuts against the easy story: the Washingtonians wrote down their own version of this rule almost immediately, circulated it in a manual and a newspaper, and dissolved anyway. Whatever preserved AA, it was not this rule on its own, because the other movement had it too.

Tradition 11, attraction rather than promotion

Technical. Load 2.69, governing four resources, with visible proof of recovery at 0.49 of its load. Principal supplier for Step Two. Degrading it alone costs 7.90 members [6.70, 9.10], the largest loss in the comparison by more than two members and the only comparison with a t statistic above 12. It is the only Tradition appearing in the arrival rate: inflow is an exogenous referral floor plus attraction generated by members' twelfth-step practice and scaled by Tradition 11. Setting it to zero leaves 13.5 members at thirty years against 41.7, with survival at 0.998. Unreferred groups, by contrast, are alive in 36.0 per cent of runs at thirty years [0.314, 0.408]. That ordering, stated as mortality rather than as size, holds in all 236 targeted perturbations and under all four structural variants. *Algebra and simulation.*

In plain terms. Attraction costs the most in the membership comparison, and the more important finding is about what it does *not* do. A group nobody has heard of becomes small and stays small, more or less forever. A group cut off from courts, treatment centres and doctors becomes mortal, because its only source of new people is the people already in it, and that can fall to zero and stay there. Being unknown makes a group small. Being cut off from the outside world kills it. That distinction is the single best-supported thing the simulation produces, and it survived every attempt made to break it.

Tradition 12, anonymity

Technical. Load 2.62, governing five resources, with confidentiality at 0.57 of its load. Principal supplier for Step Five and the Tradition governing confidentiality at the matrix maximum of 1.0. Degrading it alone costs 2.56 members [1.32, 3.80]. It is also one of two Traditions guarding the effective adherence of the group conscience, entering that term symmetrically with Tradition 9. Because the two enter identically there, the entire difference between their measured losses, 2.56 against 0.89, is Tradition 12's own governance row. In the deliberation model it is the third of the three holding maximum influence near one over N. *Algebra, deliberation and simulation.*

In plain terms. Anonymity does two separate jobs and they are easy to run together. The first is ordinary and immediate: it is what makes confidentiality available, and confidentiality is what the fifth Step needs and cannot do without. The second is structural and invisible from inside the room: by keeping anybody from becoming a name, it stops the group's decisions concentrating on one person, which is the condition for those decisions being reliable at all. Tradition 9 shares the second job and not the first, which is exactly why anonymity scores higher in the membership comparison and why that higher score says nothing about the part that matters most.

Maxwell wrote in 1950 that anonymity had "sheer survival value" and could not say why. This is the why, and the point worth keeping is that it is two whys rather than one. The Washingtonians,

for what it is worth, took the opposite position on this deliberately and with an argument, which is what makes the comparison a comparison between two written codes rather than between rules and no rules.

What the Tradition rows say taken together

Technical. The twelve split into two tiers, and the split is a property of the governance matrix rather than a reading of the text. Seven Traditions govern at least one resource some Step consumes. Five, namely 4, 6, 7, 9 and 10, govern none, so they supply nothing to any Step and appear in the simulation only as multipliers on the adherence of the Traditions they guard. That multiplier form is an assumption. The index-pairing conjecture fails on all twelve counts, and five of the twelve are arithmetic: those five Traditions have empty rows, so no sparsity-preserving perturbation can move the index-mate entry off zero. At every jitter level the proportion of draws in which pairing fails on all twelve equals the proportion in which it fails on the seven that could have gone either way, to the last draw. A threshold test that can reach the five finds them failing by margins comparable to the seven, with thresholds from 0.417 to 0.627 against a mean live governance entry of 0.374. Pairing fails on all twelve in 85.5 per cent [83.9, 87.0] of draws at plus or minus thirty per cent jitter and 40.6 per cent [38.5, 42.8] when every magnitude is randomised.

In plain terms. The Traditions come in two kinds, and the model found the division rather than being told it. Seven of them hand the group something it needs. Five of them hand over nothing and instead stop something from going wrong: they are guards rather than suppliers. Nobody sorted them that way in advance; it fell out of asking what each Tradition looks after and noticing that five of the answers were empty.

The pairing idea, that the first Step goes with the first Tradition and so on down, is wrong everywhere. But the honest version of that result is more careful than the headline. Five of the twelve failures are wrong for a boring reason: those Traditions supply nothing to anybody, so of course they do not supply their own Step. Those five are arithmetic, not evidence, and a separate test had to be built to say anything real about them. It was built, and they fail like the others. What the whole result rests on is whether the strengths in the two tables are roughly right. If they are, the pairing is dead. If a reader accepts only the pattern of which Tradition touches what and rejects every magnitude, it becomes a coin flip. So Part Four argues for its numbers rather than hiding behind a robustness percentage, and this is the one part of the book that works that way.

What this primer does not say

It does not rank the Traditions by importance. The degradation column measures membership at twenty years in a one-factor sweep from 0.85 to 0.50. Traditions 2, 9 and 12 carry the book's central argument, and three of the four unresolved rows are protective Traditions whose simulation role is an assumption rather than a derivation.

It does not establish that Traditions 2, 9 and 12 are what prevents the three obstructions. The appendix shows the three obstructions behave as the theorem says. The step from there to the claim that these three Traditions are what prevents them is a reading of three sentences, and it is the book's central claim. Nobody has a method for testing it. In plainer terms: the mathematics says what a group has to avoid, and the Traditions look very much like instructions

for avoiding exactly those things, but “looks very much like” is a judgement and no computation upgrades it.

It does not validate the two matrices. Both were built by one person. The eight resources are that person’s list and no source proposes it. A second reader marking the ninety-six governance cells is the largest outstanding item in the project and no further computation substitutes for it. Flipping four of fifty-six enabling cells at random leaves Part Four’s claims standing 86 per cent of the time; flipping sixteen leaves them at a coin flip. In plainer terms: one disagreement here and there is survivable, wholesale disagreement is not, and only a second reader can say which this would be.

Nothing here is calibrated to AA data, because none exists at the required resolution. Inflow, dropout and churn were set so that a fully adherent group holds near forty-five members with an experienced core near nine. At 400 seeds it delivers 41.7 plus or minus 1.5 members and a core of 7.7 plus or minus 0.5, undershooting both, and the shortfall on the core is larger than its own interval. That is calibration to a stylised fact, and it does not hit the stylised fact exactly.

Where every figure comes from

The five headings below are the book’s canonical reference headings, in the order `tools/check_chapter.py` requires of a chapter. The primer is exempt from that rule, because it is an appendix rather than a chapter and the checker skips its structural block entirely. It keeps the convention anyway, and the exemption is the reason a reader should not assume anything enforced it.

Read in full:

Golub, B. and M. O. Jackson (2010). “Naive Learning in Social Networks and the Wisdom of Crowds.” *American Economic Journal: Microeconomics* 2(1): 112-149. The vanishing influence condition and the three obstructions. Read at source.

Maxwell, M. A. (1950). “The Washingtonian Movement.” *Quarterly Journal of Studies on Alcohol* 11: 410-452. The claim that anonymity has “sheer survival value”. Read at source; the original text is stored in the Maxwell subdirectory of `research/incorporated/`, a retyped reproduction with four demonstrable transcription errors listed in `research/SOURCES.md`.

Grosh, A. B. comp. (1842). *Washingtonian Pocket Companion*. Second edition. Utica, N.Y.: B. S. Merrell. The Washingtonians’ own manual, used here for the written analogues of Traditions 4, 7, 9 and 10 and for their deliberate opposite position on anonymity. Read at source; saved in `research/`.

Cited at a remove:

Nothing. Every source named here was read at source. The primer restates findings from chapters that do cite at a remove, and those removes are recorded in the chapters rather than repeated here.

Internal, and reproducible from this repository:

`model/aa_group_model.py`, matrices `S` and `GOV`, the effective-adherence function, and the arrival and dropout expressions.

`model/book-calculations.ipynb`. Section 2 for the influence weights and consensus errors behind every deliberation figure; 3 for the twelve-Tradition degradation comparison; 10 for the apparatus,

including the derived group-dependence table; 11 and 11b for the coupling, the principal suppliers and the perturbation designs; 11c for the threshold test; 11d for the load column and the reassignment test; 13 for the service results behind Step Twelve; 14 for the Part Five failure modes; 17 for the resource-list test; 18 for the sparsity perturbation.

`research/tradition_paired.json`, the 400-replication paired degradation runs, produced by `model/tradition_paired.py`. `research/part5.json`, 4,800 runs behind the Tradition 3 sweep and the three failure modes. `research/ch15_service.json`, the three service configurations. `research/core_thresholds.json`, the two membership thresholds and the calibration figures. `research/resource_list.json`, the sixty-four resource-list variants. `research/oat_full.json`, the 236 targeted perturbations. `research/structural.json`, the four structural variants.

`appendix/APPENDIX.md`, sections A2 for the effective-adherence specification, A3.3 for the seed-count table, A5.4 and A5.6 for the perturbation designs and what they cannot reach, A5.4b for the threshold test, A5.4c for the reassignment test, A5.4d for the degradation ranking, A5.4e for the sparsity pricing, A9 for the structural variants and A9.5 for the resource-list test. Part Two's sensitivity analysis is the section titled "Part Two: sensitivity of the mapping between the theorem and the Traditions", which is numbered A8 and shares that number with the reproduction section at the end of the file.

Manuscript chapters 8, 9, 10, 12, 13, 15, 16, 17, 18, 19 and 20, whose Machinery sections carry the full designs and estimands for every figure quoted above.

What was not read:

The AA literature that gestures at a parallel between the Steps and the Traditions. It is AA copyright and this project does not acquire it, which means the index-pairing conjecture is refuted as a thing people believe rather than as a thing somebody published, and I cannot say how strongly the literature gestures.

No study of AA group culture, which is why the behavioural description behind Tradition 3's retention mechanism is illustration rather than evidence.

The literature on skill depreciation rates in adults, which would say whether six per cent a week is the right order of magnitude for anything comparable. It is the most easily improved number in the model and it has not been tried.